Black-Box Test Design

Introduction

This document applies black-box testing techniques to derive test cases. It uses Equivalence Partitioning and Boundary Value Analysis to ensure systematic and efficient coverage.

Equivalence Partitioning

Equivalence Partitioning divides inputs into valid and invalid groups where all values in a group are expected to behave similarly. One representative value from each partition is selected to reduce the number of tests while maintaining coverage.

Boundary Value Analysis

Boundary Value Analysis focuses on values at the edges of partitions. Typical test values include just below, at, and just above boundaries, where errors are most likely.

Mobile Phone Number

Partition	Description	Boundary	Test Values
1–7	Too short (invalid)	1,7	0,1,2,6,7
8	Valid length	8	8
9+	Too long (invalid)	9	9,10,20

Country Code

Partition	Description	Boundary	Test Values
0	Empty (invalid)	0	0
1–3	Valid	1,2,3	1,2,3
4+	Too long (invalid)	4	4,5,20

Address Components

Field	Valid	Invalid	Boundary
Address line	≥1 char	0	0,1,2

Number	≥1	0	0,1,2
City	≥1	0	0,1,2
Postal code	5 digits	<5 or >5	4,5,6
Floor	ST or 1–2	other	0,1,2,3
Door	1–50 or codes	other	0,1,50

CPR

Partition	Description	Boundary	Values
0	Empty	0	0
1–7	Too short	1,7	0,1,2,6,7
8	Valid	8	8
9+	Too long	9	9,10,20

Date of Birth

Field	Valid	Invalid	Boundary
Day	01–31	00,32	0,1,31,32
Month	01–12	00,13	0,1,12,13
Year	2 digits	other	1,2,3

Names

Partition	Valid	Invalid	Boundary
Length	≥1	0	0,1,2
Characters	Alphabetic	Numeric/special	A,1

Gender

Valid	Invalid

male	
female	
	null/other

Decision Table Example (Phone Validation)

Condition	Case 1	Case 2	Case 3
Length = 8	Yes	No	No
Numeric	Yes	Yes	No
Valid?	Yes	No	No

Reflection on Test Implementation

Although a comprehensive set of test cases was derived using black-box techniques such as equivalence partitioning and boundary value analysis, not all identified test cases were implemented.

The purpose of defining all partitions and boundary values was to ensure a systematic and thorough understanding of the input space. However, in practice, many of the identified test cases belong to the same equivalence class and are therefore expected to produce identical behavior.

For this reason, only a representative subset of test cases was implemented. This approach reduces redundancy while still maintaining effective test coverage. Implementing every possible test case within the same partition would not provide additional value but would increase maintenance effort and reduce clarity.

Furthermore, some edge cases were considered less relevant in the given context, either due to system constraints or because they fall outside the intended use of the application.

To conclude: we decided not to test everything, but to test what we felt was effectively and efficiently.

White-Box Testing

Coverage | unit in fake-info

Element	Class, %	Method, %	Line, %	Branch, %
all	52% (9/17)	61% (32/52)	76% (123/161)	66% (32/48)
dk.fakeinfo	0% (0/2)	0% (0/3)	0% (0/14)	0% (0/10)
dk.fakeinfo.controller	0% (0/2)	0% (0/12)	0% (0/17)	0% (0/2)
dk.fakeinfo.model	60% (6/10)	61% (8/13)	66% (10/15)	100% (0/0)
dk.fakeinfo.repository	100% (0/0)	100% (0/0)	100% (0/0)	100% (0/0)
dk.fakeinfo.service	100% (3/3)	100% (24/24)	98% (113/115)	88% (32/36)
FakePersonService	100% (2/2)	100% (20/20)	100% (105/105)	96% (31/32)
PersonNameService	100% (1/1)	100% (4/4)	80% (8/10)	25% (1/4)

Static testing tools

src/test/java/unit/FakePersonServiceTest.java

Remove this block of code, fill it in, or add a comment explaining why it is empty. Intentionality
Maintainability Medium suspicious +
Open Not assigned L116 • 5min effort • 3 days ago • Code Smell • Major

Use isZero() instead. Intentionality
Maintainability Low assertj tests +
Open Not assigned L248 • 5min effort • 3 days ago • Code Smell • Minor

Use isZero() instead. Intentionality
Maintainability Low assertj tests +
Open Not assigned L794 • 5min effort • 3 days ago • Code Smell • Minor

Use isZero() instead. Intentionality
Maintainability Low assertj tests +
Open Not assigned L843 • 5min effort • 3 days ago • Code Smell • Minor